Mohit

themohitrao02@gmail.com | linkedin.com/in/mohit-yadav-a56295252

Education

The Ohio State University **2026 - Present**
PhD in Reliable Reasoning and Self-Improving AI Agents, Dept. of Computer Science and Engineering
Current GPA: 3.89 / 4.00

IIT DELHI **2021 - 2026**
B.Tech & M.Tech (Dual Degree) - Mechanical Engineering
CGPA: 8.94 / 10

Experience

AI Intern | Mastercard AI Garage **May 2025 - Jul 2025**
Advanced Merchant Identity Fraud Detection using Siamese Networks
- Patent: Invented and developed novel card-based merchant features, leading to a 20% increase in model accuracy
- Spearheaded detection of merchants engaging in fraudulent identity reporting on a high-dimensional, real-world Mastercard dataset
- Crafted merchant and card embeddings via Siamese Networks, integrated with XGBoost for 100-class industry classification
- Developed merchant embeddings with the TabNet pre-trainer, enhancing card features integrated with XGBoost for classification

AI Evaluation Author | AfterQuery **Jul 2025 - Present**
- Authored and validated 23/23 accepted Terminal-Bench tasks spanning C, Rust, Go, Python, and WASM, each shipping a calibrated reference solution passing 1- 6 of 8 frontier-model runs
- Submitted 16 vulnerability-reproduction tasks on Project Fenrir, working across multi-KLOC C, Rust, and JVM repositories
- Promoted from Expert to Reviewer on Project Julio, grading and validating other contributors' submitted tasks

Software Intern (AI Project) | Clintico Infotech Pvt. Ltd. **Sept 2025 - Nov 2025**
Multi-agent architecture pipeline: Reconstruction, Validation, Prompt, SQL-Generation and Self-Refinement
- Built a multi-agent system integrating a SQL database, bypassing token limits and reducing LLM costs by 40% via data preprocessing
- Improved SQL column identification accuracy to 95% by deploying Forest of Thoughts (FoT) over the Few-Shot CoT reasoning framework
- Built an automated SQL agent for dynamic query code generation, enabling what-if analysis and eliminating manual retrieval

NLP Intern | Dyota.ai **Dec 2023 - Feb 2024**
Hospitality chatbot deployed on WhatsApp
- Built a hospitality chatbot answering user queries with an open-source LLM and deployed it on WhatsApp
- Preprocessed text data using stemming, lemmatizing and stop-word removal
- Fine-tuned Llama2 using LoRA, improving understanding and response generation for hospitality queries
- Implemented RAG with Pinecone for efficient retrieval and integrated conversational memor
- Deployed on WhatsApp using Twilio, AWS EC2, ngrok and Flask to build the full messaging stack

Data Science Intern | CandiBrain **Nov 2022 - Mar 2023**
AI-Powered Stock Market Prediction and Trading System
- Developed a stock market bot using AI to forecast trends and autonomously execute trading decisions
- Built an LSTM time series model to predict Infosys stock prices after preprocessing and feature engineering
- Trained a model-free DQN reinforcement learning agent to take actions on current derivative (options) prices
- Used Kite Developer Tools for historical data, WebSockets for live data streaming, and the API to place orders

Projects

Physics-Informed Neural Network and Flood Assessment | Prof. Anand Abraham **Jul 2025 - Apr2026**
- Developed an integrated PINN + Sentinel-1 framework for flow forecasting, flood inundation assessment in the Mahanadi basin
- Designed the PINN using 9 stations and 2006-2020 data, integrating mass balance flow forecasts outperformed ML and DL models
- Implemented a physics loss, improving consistency by 25% and achieving 20% better generalization vs. ARIMA/LSTM baselines
- Detected floods at 25,015 m^3/susing the rating curve, achieving NSE = 0.82 (Random Forest), with PCA for pattern identification

Codexia: AI Tutor for Future Coders | Prof. Sumeet Agarwal, Prof. Kushal Shah **Sept 2024 - Nov 2024**
- Conducted a comparative analysis of open-source and proprietary NLP models with a focus on advanced code generation
- Authored and published research paper, *LLMs as Coding Mentors: Evaluating Effectiveness in Python Instruction*, at ACM conference
- Developed a GenAI code tutor using CodeLlama 70B, with a custom synthetic dataset from GPT-4 and fine-tuning using QLoRA

Sentiment Analysis of IMDb Movie Reviews | NLP and Neural Networks **May 2024 - Aug 2024**
- Preprocessed text via tokenization, stop-word removal and lemmatization
- Applied Word2Vec for feature extraction with logistic regression for classification, achieving 85% accuracy
- Built a deep neural network with the TensorFlow Keras API, improving sentiment prediction accuracy to 92%

Technical Skills

Machine Learning: XGBoost, Random Forest, LSTM, DQN (Reinforcement Learning), Siamese Networks, TabNet, PCA, Physics-Informed Neural Networks, TensorFlow/Keras
LLMs & NLP: Llama2, CodeLlama, LoRA/QLoRA fine-tuning, RAG, Pinecone, Chain-of-Thought / Forest of Thoughts, Word2Vec, multi-agent pipelines
Data & Deployment: SQL, Flask, Twilio, AWS EC2, ngrok, WebSockets, Kite API, Sentinel-1 SAR processing